

Baseline Model

This section explores the pretrained embeddings produced by the DINOv2 ViT-S/14 model. DINOv2 was selected because its self-supervised training produces general-purpose visual representations without requiring task-specific labels, making it well suited to the unsupervised setting considered in this work. The ViT-S/14 (approximately 21 million parameters) variant was chosen as a relatively compact model that provides both global CLS and local patch representations, allowing the same backbone to support the image-level experiments in this study and subsequent investigation of local anomaly representations.

The purpose of this section is to:

1. Evaluate several anomaly scoring methods using the extracted CLS embeddings:
 - Centroid Distance
 - K-Nearest Neighbours (KNN)
 - Mahalanobis Distance
2. Analyse the distribution of anomaly scores for normal and defective samples to determine how well each method separates the two classes.
3. Compare methods using AUROC to identify the most effective baseline anomaly scoring approach. AUROC is used because it evaluates ranking performance across all possible decision thresholds, avoiding the need to select a fixed anomaly threshold for baseline comparison.
4. Identify categories that are already well separated within the embedding space, as well as categories that remain challenging and may benefit from additional experimentation.

The results obtained in this section serve as a baseline against which subsequent approaches, including fine-tuned and semi-supervised representations, are compared.

1. Evaluate several anomaly scoring methods using the extracted CLS embeddings

Metrics chosen include:

- **Centroid Distance:** Computes the Euclidean distance between each test embedding and the centroid (mean embedding) of the corresponding training category.
- **KNN:** The training embeddings are used as the reference set for KNN. Each test embedding is then compared against them to find its K nearest training embeddings and their distances. The anomaly score is taken as the distance to the closest neighbour.
- **Average KNN:** Computes the average distance between each test embedding and its K (K=5) nearest training embeddings.
- **Mahalanobis Distance:** Computes the distance between each test embedding and the training distribution while accounting for the covariance structure of the embedding space.

2. Analyse the distribution of anomaly scores for normal and defective samples

A reusable plotting function is used to compare the anomaly score distributions of normal and defective test samples for each category. The score distributions are plotted separately for each anomaly scoring method, allowing the degree of separation between normal and defective samples to be visually inspected.

AUROC is also computed for each category to quantify how well the selected scoring method ranks defective samples above normal samples.

Method	mean	median	std	Min Category	Value	Max Category	Value
Centroid	0.845	0.849	0.135	screw	0.506	leather	0.997
KNN	0.923	0.933	0.06	screw	0.804	leather	1.0
Avg-KNN	0.92	0.926	0.062	screw	0.786	leather	1.0
Mahalanobis	0.958	0.972	0.049	screw	0.802	bottle	1.0

AUROCS for each scoring method

The centroid-based method has reasonable performance across several categories; however, there is a noticeable overlap between the score distributions of normal and defective samples. This indicates that the distance from the centroid alone is not enough to fully describe the embedding space. While the centroid gets the average location of normal samples, it does not take into account the spread or covariance of the distribution, meaning some anomalous samples receive scores similar to normal samples.

The K-Nearest Neighbour approach improves upon the centroid-based method by producing clearer separation between the score distributions of normal and defective samples. The leather category has perfect separation, achieving an AUROC of 1.0. This suggests that anomalous samples are characterised more by their distance to nearby normal samples than by their distance to a single global centroid. The results indicate that normal and defective samples occupy distinct regions of the embedding space, allowing KNN to identify anomalies through local neighbourhood structure rather than relying solely on the overall centre of the distribution.

Average KNN does not improve upon the performance of standard KNN and achieves a lower average AUROC across the dataset. This suggests that the nearest normal neighbour contains the most informative signal for anomaly detection. By averaging across multiple neighbours, the anomaly score becomes influenced by additional normal samples that may lie further away in the embedding space, reducing the contrast between normal and defective samples. As a result, the score distributions exhibit greater overlap, leading to weaker separation and reduced anomaly detection performance.

The Mahalanobis score performed best among all evaluated methods, achieving perfect separation between normal and defective samples in several categories, including bottle, leather, and tile. Mahalanobis achieved the highest mean AUROC overall. KNN performed marginally better on screw, achieving 0.804 compared with 0.802 for Mahalanobis. Overall, Mahalanobis was therefore selected as the preferred anomaly scoring method for this study.. Furthermore, the categories exhibiting perfect separation could be used as control groups in future experiments to evaluate whether proposed modifications improve anomaly detection performance or instead degrade the structure and separability of the embedding space.

Despite strong overall performance, categories such as screw achieved lower AUROC values across multiple scoring methods. This suggests that normal and defective samples occupy more similar regions of the embedding space, making anomaly detection more challenging. These categories provide useful benchmarks for evaluating future improvements, as successful methods should increase separability in these difficult cases.